

MoveMate: Supporting Video-Guided Workouts through Motion Visual Analysis and Multi-Device Guidance

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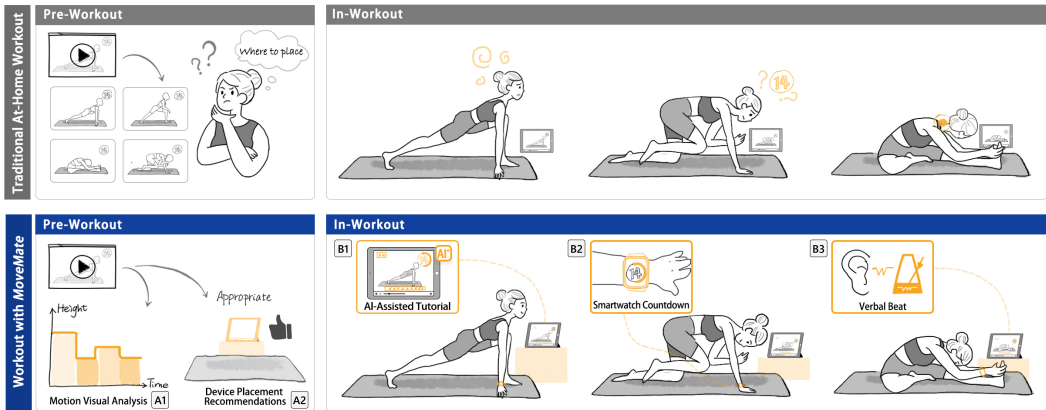


Fig. 1. Traditional video-guided workouts (top) versus workouts supported by *MoveMate* (bottom). *MoveMate* enhances video-guided workouts through support across the pre-workout and in-workout phases. Before the workout, the tablet provides visualizations of the instructor's face orientation and eye height changes (A₁), along with device placement recommendations (A₂). During the workout, the tablet delivers AI-assisted tutorials (B₁) and rhythmic audio cues (B₂), while the smartwatch provides countdown support (B₃).

Online instructional workout videos often involve movements for different muscle groups, leading to frequent transitions between different body postures, such as prone, supine, and standing postures, which repeatedly shift the user's viewing direction. However, users often place playback devices in a fixed location, so they may struggle to see the screen during certain movements, leading to frustration and an increased risk of injury. To mitigate this issue, we present *MoveMate*, a multi-device workout support system. Before the workout, *MoveMate* analyzes the instructional video, visualizes changes in the instructor's face orientation and eye height, and provides device placement recommendations. During the workout, the system delivers additional visual and auditory guidance, helping users follow movements across different body postures. Our user study

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($N = 24$) showed that *MoveMate* improves users' experience of following the instructor under reduced visibility. Our findings and reflections offer insights for designing future computer-assisted workout systems.

CCS Concepts: • **Applied computing** → Health care information systems; • **Human-centered computing** → **Ubiquitous and mobile computing systems and tools**; **User studies**.

Additional Key Words and Phrases: Mobile Device, Video-Guided Workout, Smartwatch, Multi-Device System

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1 Introduction

The recent surge in at-home workouts, driven by modern fast-paced lifestyles and accelerated by the COVID-19 pandemic, has led to an increasing reliance on video-guided instruction [2, 17, 37, 64]. While fitness platforms (e.g., 8Fit¹) and video-sharing websites (e.g., YouTube²) offer a vast array of instructional content, an important challenge is rising. Many instructional workout videos include movements performed in different postures, such as prone, supine, and standing postures. These postural changes can substantially alter users' line of sight (see Figure 2). Because such movements are often interleaved throughout a workout, users' viewing conditions may change repeatedly during a single session. However, users typically place their playback device in a fixed location, which can leave them with poor viewing angles for many movements. As a result, users may instinctively twist or crane their bodies to regain visual guidance, especially if they are beginners unfamiliar with the movements. Such compensatory postures can disrupt body alignment and stability, thereby increasing the risk of muscle strain or joint injury [16]. Furthermore, repeated visual disconnections can lead to frustration, reduced focus, and decreased motivation, ultimately degrading the quality of the workout experience.

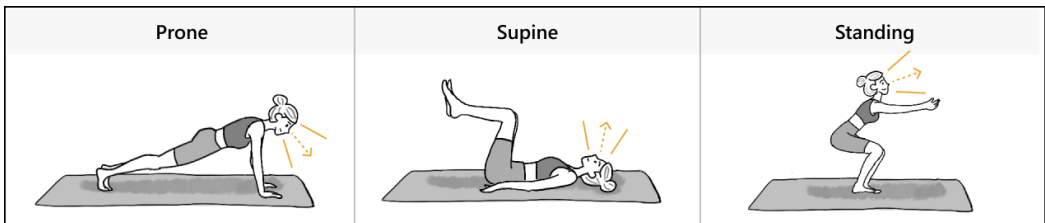


Fig. 2. Three typical fitness postures: prone, supine, and standing in workout instructional videos.

Prior work has explored several strategies to mitigate this screen visibility issue in video-based workouts. Some research has employed robot-mounted displays [46] or autonomous furniture [11] to dynamically adjust the screen's placement, keeping it within the user's field of view. Another approach [35, 66] replaces traditional screens with head-mounted displays (HMDs) to ensure uninterrupted visual access. However, these approaches typically constrain adaptation to either horizontal repositioning [46] or vertical height adjustment [11], and some may be less suitable for fast-paced [11] or floor-contact movements [35, 66]. Moreover, they often depend on expensive, specialized, and bulky hardware, reducing portability and making them less practical for everyday or travel use.

¹8fit.com

²youtube.com

In this paper, we introduce *MoveMate*, a portable, enhanced fitness workout system designed to help users follow video exercises without losing track of instructions when users often cannot see the screen. *MoveMate* addresses this using everyday portable devices—tablets and smartwatches—making it easy to workout anywhere, whether at home, in a hotel, or in a school dance studio. Our system enhances the video-guided workouts through two phases: 1) Before the workout, *MoveMate* provides visualization of the instructor’s face orientation and eye height changes and device placement recommendations based on the specific movements in the instructional video, ensuring the screen remains as visible as possible throughout the workout; 2) During the workout, the system delivers additional visual and auditory guidance to compensate for moments when the user cannot see the screen. Through a user study with 24 participants, we demonstrate that *MoveMate* effectively helps users follow video-guided workouts when the screen is difficult to see.

The main contributions of this paper are as follows:

- (1) The identification and characterization of user requirements that arise when visual access is constrained during video-guided workouts.
- (2) The design and implementation of *MoveMate*, a novel video-guided workout system that supports users through motion visual analysis and multi-device guidance.
- (3) A user study that evaluates the effectiveness of *MoveMate* and distills reflections for future computer-assisted workout systems.

2 Related Work

To ground our work, we first broadly reviewed prior computer-assisted physical activity systems, and then examined existing strategies for addressing the specific visual challenges of video-based workouts in depth.

2.1 Computer-Assisted Physical Activity Systems

A substantial body of research (see Table 1) has explored physical activity systems that leverage computational techniques to support learning and practice. These systems can be categorized based on the timing of guidance: before, during, or after the activity.

Some systems provide pre-workout guidance, which means that users will receive some guidance from the system before they start exercising. This is particularly beneficial for novices learning complex movements. The dominant modality in this phase is visual, often involving previews of the ideal movement augmented with graphical overlays like skeletal joints [66] or motion trajectories [13, 41]. By clearly establishing the task requirements upfront, these systems set a mental model for correct execution.

Most systems provide in-workout guidance, which implies that users will receive real-time feedback or suggestions during the actual course. At this stage, the guidance provided often involves multi-modal information. For example, these can be: auditory (sound), visual or haptic (vibration). Visual cues remain central, especially in video-based training where users imitate on-screen instructions [56, 66]. Systems often use real-time overlays such as skeletal visualizations [7, 59, 66], directional indicators ► [56], and alignment markers  [56] to highlight deviations from the correct form. To complement vision, auditory cues—ranging from direct, low-cognitive-load instructions [67] to sonification techniques that encode performance errors in auditory parameters like pitch [54] or timbre [61]—have also been widely explored to support non-disruptive and seamless user feedback. Vibration feedback is also present in some systems, where information is often transmitted through wearable devices. Compared to audio, it can convey messages more silently or privately [8]. Vibration feedback has been proven to effectively correct posture [65] or maintain balance [3], particularly in situations of visual or audio overload.

Table 1. Guidance in computer-assisted physical activity systems, categorized by delivery modality and timing.

Citation	Physical Activity	Modality			Phase		
		Visual	Auditory	Haptic	Pre-Workout	In-Workout	Post-Workout
Tajadura-Jimenez et al. [57]	General Fitness Exercise	☐	☒	☒	☐	☒	☐
Yu et al. [71]	General Fitness Exercise	☒	☐	☐	☐	☒	☐
Xia et al. [67]	General Fitness Exercise	☒	☒	☐	☒	☒	☒
Wu et al. [66]	General Fitness Exercise	☒	☐	☐	☒	☐	☐
Semeraro & Turmo Vidal [56]	General Fitness Exercise	☒	☐	☐	☐	☒	☐
Clarke et al. [14]	General Fitness Exercise	☒	☒	☐	☒	☒	☒
Pham et al. [49]	General Fitness Exercise	☒	☐	☐	☐	☒	☐
Fieraru et al. [21]	General Fitness Exercise	☒	☐	☐	☐	☒	☐
Kim et al. [36]	Resistance Training	☒	☒	☐	☐	☒	☐
Rabbi et al. [51]	Resistance Training	☒	☐	☐	☐	☒	☐
Wang & Ma [63]	Physical Therapy	☒	☐	☐	☐	☒	☐
Tang et al. [58]	Physical Therapy	☒	☐	☐	☐	☒	☐
Xia et al. [68]	Limb Exercise	☐	☒	☐	☐	☒	☐
Tharatipyakul et al. [59]	Tai-Chi	☒	☐	☐	☐	☒	☐
Han et al. [24]	Tai-Chi	☒	☐	☐	☐	☒	☐
Lin et al. [41]	Basketball	☒	☐	☐	☒	☒	☒
Cannavò et al. [7]	Basketball	☒	☐	☐	☐	☒	☒
Cheng et al. [13]	Basketball	☒	☐	☐	☒	☒	☒
Scrucca & Karlis [55]	Basketball	☒	☐	☐	☒	☐	☒
Chan et al. [10]	Dancing	☒	☐	☐	☒	☒	☒
Turmo Vidal et al. [61]	Dancing	☐	☒	☒	☐	☒	☐
Großhauser et al. [23]	Dancing	☐	☒	☐	☐	☒	☐
Schaffert & Mattesc [54]	Rowing	☐	☒	☐	☐	☒	☐
Iannucci et al. [30]	Rowing	☒	☐	☐	☐	☒	☐
Ma et al. [44]	Table Tennis	☒	☐	☐	☒	☒	☐
Newbold et al. [48]	Stretching	☐	☒	☐	☐	☒	☐
Liu et al. [42]	Running	☒	☐	☐	☐	☐	☒
Daniel et al. [9]	Swimming	☐	☒	☐	☐	☒	☐
Berentsen et al. [3]	Mountain Biking	☒	☐	☒	☐	☒	☒
Woźniak et al. [65]	Golf	☒	☒	☒	☐	☒	☐
Turmo Vidal et al. [60]	Yoga, Circus & Weightlifting	☒	☒	☒	☐	☒	☐
Duente et al. [18]	Cycling	☐	☐	☒	☐	☒	☐
Yoo & Kay [70]	Exergame	☒	☐	☐	☐	☐	☒

In addition to this, some systems focus more on post-training feedback, such as summaries of athletic performance or opportunities for reflection. Such feedback is often presented visually through simple numerical values, visualizations [10, 14, 42], or slow-motion replays [10, 14]. This information can be particularly valuable for users seeking to review and improve their athletic behavior and performance.

2.2 Strategies for Maintaining Visual Continuity in Video-Based Workouts

We have found that in addition to the computer-assisted physical activity systems mentioned above, there are several systems addressing the same scenario as ours: exercising at home by following video content. They also recognized a similar issue to the one we identified regarding video-guided workouts: the difficulty of clearly seeing instructions due to shifts in the line of sight and physical

Table 2. Comparison of systems addressing screen invisibility during video-based workout.

Citation	Methods	Supports Horizontal Visibility?	Supports Vertical Visibility?	Supports Fast-Paced Movements?	Supports Floor-Contact Movements?
Chen et al. [11]	Automated screen height adjustment via a retrofitted commercial lift desk.	✗	✓	✗	✓
Nakamura et al. [46]	Robot-mounted screen with rapid omnidirectional mobility that mimics the video instructor's movements.	✓	✗	?	?
Jo et al. [35]	Replace traditional screen with an AR HMD.	✓	✓	?	✗
Wu et al. [66]	Replace traditional screen with an AR HMD.	✓	✓	✓	✗
<i>MoveMate</i>	Pre-workout motion analysis and device placement recommendations & In-workout multi-device guidance.	✓	✓	✓	✓

movement during exercise. Some studies tackle this issue through dynamic screen positioning. For instance, researchers have modified and custom-designed hardware systems to create adaptive environments, such as retrofitting commercial lift desks for automated height adjustment [11] or integrating projectors onto mobile robot bases to create follow-me displays that mimic the spatial movements of an instructor [46]. While effective, these solutions rely on specialized, intrusive, and space-consuming hardware setups, making them difficult to integrate into home exercise routines, limiting their scalability for everyday users, and preventing their deployment in mobile or travel contexts. Moreover, these approaches typically constrain adaptation to a single axis—lift-desk-based setups can adjust screen height but remain fixed in the horizontal plane [11], whereas robot-mounted screens can reposition around the user horizontally but remain at a fixed height, limiting their ability to accommodate vertical changes [46]. Such single-axis adaptation mechanisms can compensate for only one direction of viewpoint change and therefore struggle to accommodate the compound shifts in viewpoint that commonly arise in real-world video-guided workouts. In addition, these systems may be particularly limited in workouts involving fast-paced movements (e.g., high knees), where users' viewpoint can change quickly and substantially. Chen et al. [11] reported that when workout tempo is high, or when shoulder height changes frequently with large amplitude, their automated height-adjustment algorithm may be insufficient to help users keep up with the exercise pace.

Some other studies address these visual challenges through immersive technologies such as Augmented Reality (AR) and Virtual Reality (VR). For example, Jo et al. [35] and Wu et al. [66] used HMDs to overlay guidance directly onto the user's view, ensuring constant visibility. However, HMD-based systems may be less suitable for fast-paced or floor-contact movements. For example, Jo et al.'s AR system [35] primarily targets slow routines such as yoga, and its applicability does not necessarily extend to exercises involving jumping or rapid turns. Wu et al. [66] further reported that the rear hardware module and overall weight of the HMD made supine movements difficult, sometimes requiring participants to adjust their posture to avoid contact between the device and the ground. In addition, HMDs also pose notable comfort, safety, and accessibility challenges for workout use. They are often bulky [12] and uncomfortable during prolonged physical activity [52], can induce cybersickness [39] or neck strain [33], and remain considerably more expensive and less portable than common household devices such as tablets.

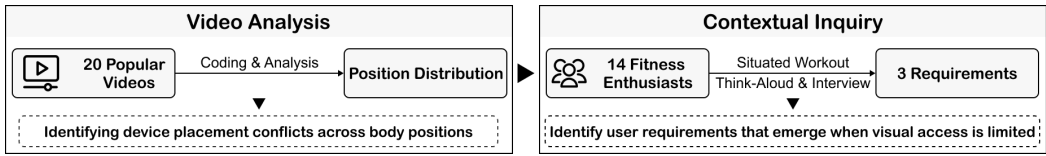


Fig. 3. The two phases of our formative study: video analysis (Left) and contextual inquiry (right).

Therefore, existing work still lacks practical and scalable solutions that address visual discontinuity without relying on specialized equipment, while supporting screen visibility in both horizontal and vertical dimensions and across diverse movement types. We introduce *MoveMate*, a system designed to support video-guided workouts using widely available commercial mobile devices and through motion visual analysis and multi-device guidance.

3 Formative Study

To align our system design with actual user needs, we conducted a two-phase formative study (Figure 3). First, we systematically analyzed online workout videos to understand the distribution of postures across various movements. This was followed by a contextual inquiry to elicit user requirements, specifically focusing on challenges that arise when the screen is difficult to view.

3.1 Formative Study 1: Analysis of Public Workout Videos

We analyzed 20 widely viewed YouTube workout videos (each with more than 10 million views; see details in our supplementary material). Two co-authors first worked together to segment each video into discrete clips, with each clip representing a single continuous exercise set (e.g., 30 seconds of push-ups). The clips were then classified into seven categories based on an established positional taxonomy from exercise science [38]: *supine*, *prone*, *side-lying*, *quadruped*, *kneeling*, *seated*, and *standing*. To ensure reliable classification, the two coders first independently classified a subset of the data ($N = 80$ clips, approximately 20% of the dataset), achieving strong inter-rater reliability (Cohen's kappa, $\kappa = 0.89$) [15, 45]. They then discussed the discrepant cases and reached consensus on their classification. The remaining clips were subsequently split evenly between the two coders and classified individually following the agreed classification scheme. Finally, we got 396 movements totally, revealing that *supine* ($N = 115$), *prone* ($N = 108$), and *standing* ($N = 108$) were the three most common postures, while others were comparatively rare (see Table 3 for details). Every analyzed video featured more than three distinct posture types and at least three instances of posture transitions, indicating that a user's gaze direction fluctuates significantly in both vertical elevation and horizontal orientation throughout a single workout.

This diversity creates inherent spatial conflicts for optimal device placement:

- **Vertical Disparity:** While *standing* exercises necessitate that the screen be elevated to approximately eye level for ergonomic viewing, *supine* (lying face-up) and *prone* (lying face-down) postures shift the optimal viewing plane to the ground level.
- **Horizontal Divergence:** Optimal screen placement also varies across the mat's axes. For *prone* movements, positioning the device at the short edge of the mat (near the head) provides the most direct line of sight. However, *supine* movements often involve body rotations or lateral orientations, making a long-edge placement more effective.

Therefore, if users do not constantly adjust the position of their playback devices, they often have to sacrifice movement accuracy to keep the screen in view. To observe the reality of home workouts, we implemented a contextual inquiry. We examined how users exercise while following workout videos, including: 1) How they position their devices; 2) Whether they frequently adjust these positions; 3) Whether they encounter situations where the screen is invisible or requires

Table 3. Posture distribution of movements from 20 public workout videos. The first column lists the video ID; subsequent columns show each movement in the video sequence, encoded with an icon for its posture type: *Supine* , *Prone* , *Standing* , *Side-lying* , *Quadruped* , *Kneeling* , and *Seated* . Icons are also color-coded by movement dynamics: black indicates **dynamic** movements, whereas gray indicates **static** movements.

NO. Movement Sequence	
1	                            

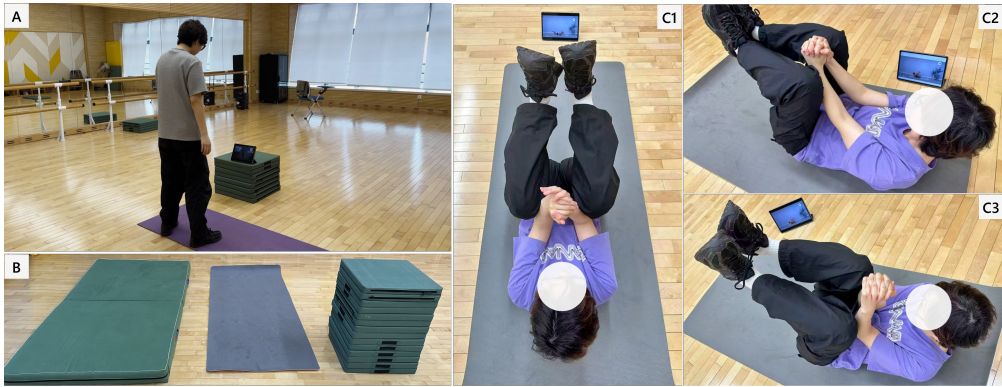


Fig. 4. Contextual Inquiry On-site Photos: (A) On-site environment. (B) Tools provided in the experimental setup. (C) Three representative initial tablet placement positions: (C₁) Front of Feet; (C₂) Head-Side; and (C₃) Foot-Side.

playback device—a Vivo Pad Air tablet [62] after previewing the workout video. They were also free to adjust the tablet’s placement at any time during the workout. Participants were instructed to use a Think-Aloud protocol [19] to verbalize their thoughts and feelings during the workout. Our two study facilitators observed and recorded each session, documenting participants’ tablet placement throughout the workout as well as any verbalized thoughts.

- (3) **Semi-Structured Interview:** Following the workout, we conducted semi-structured interviews to gain deeper insights into their experience, focusing on moments of difficulty or frustration. To derive the design requirements (DRs), we transcribed the interview data and triangulated the transcripts with participants’ recorded behaviors and think-aloud observations. Two co-authors first independently reviewed these materials through an open coding process to identify recurring challenges, behavioral patterns, and unmet needs. They then compared and discussed their interpretations, iteratively consolidating related observations into higher-level design concerns, which served as the foundation for our four final DRs.

DR1: Optimizing Global Visual Access. Our observations revealed that the majority of participants (10/14) positioned themselves based solely on the first movement. They did not quickly scan the entire sequence to choose their location. Furthermore, once the workout began, 11 out of 14 participants never adjusted their position. Consequently, they were unable to clearly see the instructor during many of the subsequent movements. This finding indicates that users require guidance or a quick way to understand changes in their visibility during this activity. By providing this information, they can significantly avoid the numerous line-of-sight obstructions they would otherwise encounter during the workout.

DR2: Rhythmic Guidance on Dynamic Movements. Most participants (9/14) expressed a strong desire to keep pace with the on-screen instructor, saying that moving in rhythm made the workout feel more motivating and gave them a stronger sense of achievement. As E11 stated, “*I wanted every repetition to match the instructor’s timing exactly. Staying perfectly on beat gave me a great sense of accomplishment.*” This need was particularly pronounced during dynamic movements, where tempo is essential for correct execution and maintaining workout flow. When visual contact was lost, participants also lost their primary rhythmic reference and often became unsettled, actively attempting to re-establish visual contact. This effort added physical strain, increased movement errors, and ended up disrupting their

rhythm even more. These findings suggest that users need guidance to help them establish and maintain the correct exercise rhythm.

DR3: Enhanced Guidance on Complex Movements. More than half of the participants (8/14) said they needed extra guidance when a movement was hard to perform and they could not clearly see the screen. In these movements, their primary learning strategy—real-time visual imitation—was missing. This forced them to disrupt their workout rhythm, often pausing for several beats to decipher the movement. As E5 noted, “...I couldn’t see the screen... and I wanted to know in advance how arms and legs I should move.” However, experienced participants did not share this concern, with two university physical education instructors (E13 and E14) noting that they could understand new movements quickly even with limited visual cues. This contrast suggests that for non-expert users, the combination of movement complexity and visual occlusion creates a significant cognitive barrier. They require supplemental guidance—beyond the standard brief visual demonstration—to bridge this gap and maintain workout continuity.

DR4: Temporal Awareness on Static Movements. Five participants emphasized the importance of knowing the remaining time for each movement. The absence of the on-screen countdown timer led to what E8 described as “temporal uncertainty” and anxiety: “During M4, I couldn’t see the on-screen countdown at all, which made me feel very anxious. I wish there were an alternative way to show how much time was left.” This need was especially evident during static postures, where users had to hold a posture without access to visual cues. For instance, during the *Straddle Forward Fold*—a static prone posture in which the participant leans forward with their face oriented completely toward the floor—visual access to the screen was entirely lost. Participants reported that, in such postures, the lack of temporal feedback heightened anxiety. These experiences suggest a clear need for an ambient channel to track duration during static movements.

While our analysis also identified other needs in video-guided workouts (e.g., heart rate monitoring, real-time performance feedback, motivational encouragement, and calorie tracking), we deliberately scoped our design to address the four core requirements directly stemming from visual occlusion. These broader needs, though valuable, fall outside the primary focus of this work.

4 MoveMate

Based on the insights from our formative study, we designed and implemented *MoveMate* to address the four design requirements and enhance video-guided workouts through two phases: (1) **Pre-Workout Phase:** *MoveMate* analyzes the workout video in advance and provides visualizations of the instructor’s face orientation and eye height over time, together with tailored device placement recommendations (DR1); (2) **In-Workout Phase:** *MoveMate* delivers multi-device guidance to support users during moments when the main screen is difficult or uncomfortable to view. It includes three design components: Audio-Visual Tutorial (DR3), Verbal Beat (DR2), and Smartwatch Countdown (DR4).

4.1 Phase 1: Pre-Workout Analysis and Preparation

Before the workout, *MoveMate* first generates a Motion Visual Analysis (see Figure 5), which features a multi-view temporal visualization (DR1) to help users better prepare for the upcoming session.

- **A timeline-based line chart** (see Figure 5A) tracks two critical metrics that dictate the user’s visual trajectory: *eye height* (corresponding to the vertical field of view, FOV) and *face orientation* (reflecting the horizontal FOV). The x-axis represents the video frame index, while the y-axis denotes the eye height.

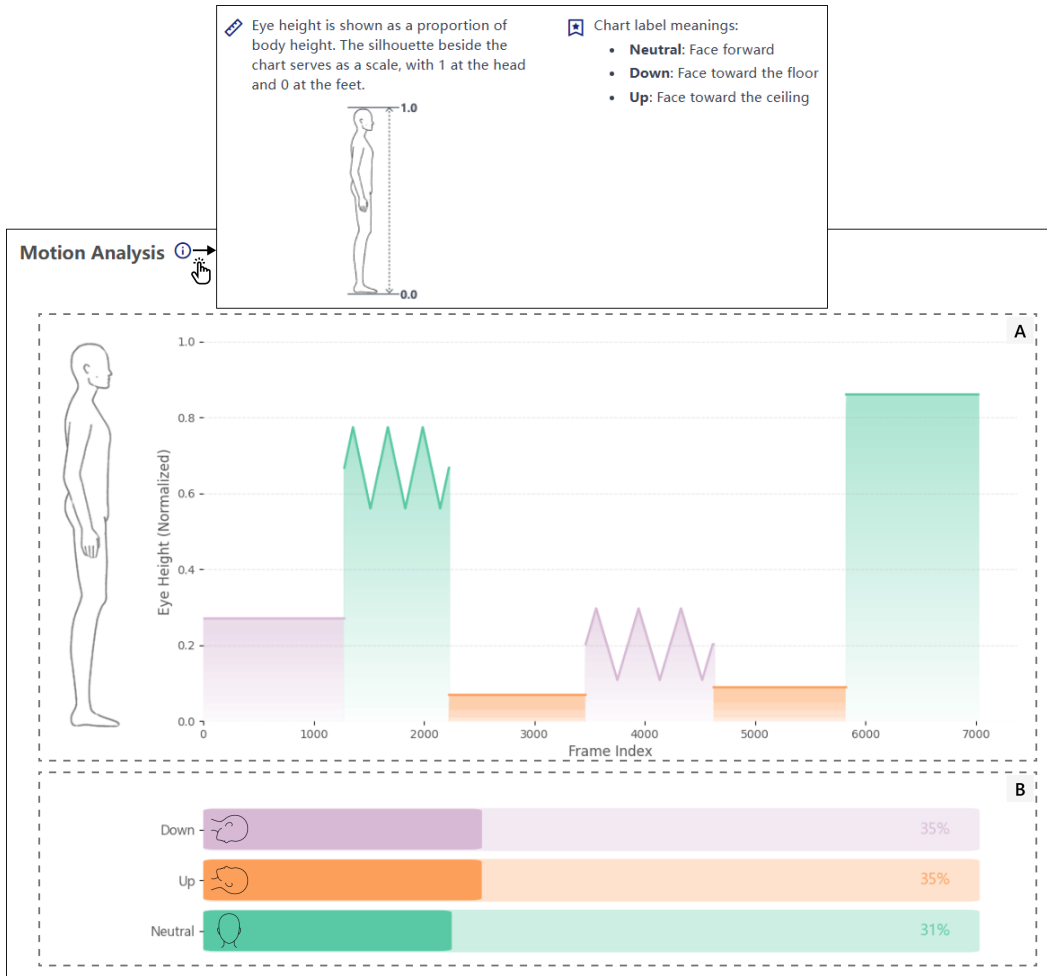


Fig. 5. Example Motion Visual Analysis. (A) The instructor's normalized eye height over time (x-axis: video frame index), segmented into color-coded intervals for face orientation. (B) The proportion of video time spent in each orientation using the same color scheme.

- **A horizontal bar chart** (see Figure 5B) aggregates the total duration of each orientation, offering a high-level “visual footprint” of the routine for quick reference.
- **Categorical color-coding** (Neutral, Up, and Down) is applied to *face orientation* to highlight shifts in attention and positional variation over time. This color-coding scheme is consistently applied to both the timeline-based line chart and the horizontal bar chart. In the line chart, the data is segmented into color-coded intervals, with each color representing a specific *face orientation* that matches the corresponding labels in the horizontal bar chart.

From Motion Visual Analysis, users can evaluate a workout's compatibility with their physical readiness or environmental constraints at a glance. For instance, a user seeking a standing-only routine can prioritize videos by identifying profiles where *eye height* remains consistently elevated. This preemptive overview not only manages expectations regarding the workout's structure (e.g.,

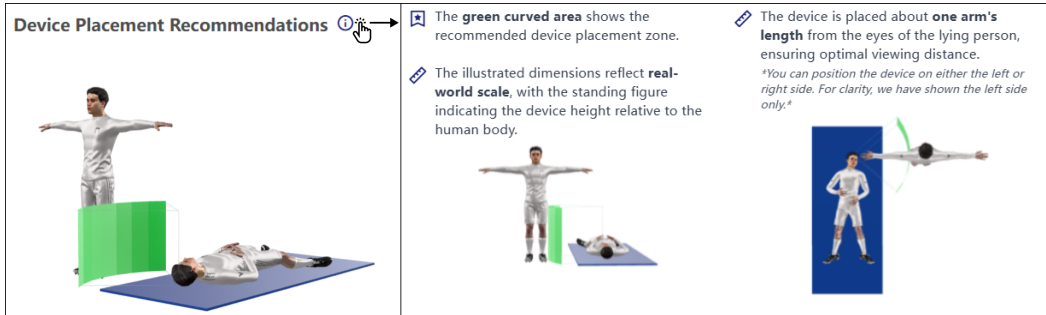


Fig. 6. Example Device Placement Recommendations. The interactive 3D model allows users to rotate 360° and zoom in/out to explore the recommended setup from multiple viewpoints.

time spent on the floor vs. standing) but also mitigates anticipatory anxiety, thereby fostering sustained motivation.

To further support user readiness, the system offers Device Placement Recommendations (see Figure 6), providing actionable guidance for positioning the playback device optimally (**DR1**). This guidance is delivered via an interactive 3D model that users can zoom and rotate to explore the suggested placement region. Our system recognizes and analyzes all movements included in the workout video and computes a fixed spatial region representing a “globally optimal” placement averaged across all planned movements. We do not provide real-time calculations to determine the optimal placement for every individual movement. Instead, we provide a globally optimal position that accommodates the majority of actions. This approach considers user requirements and preferences. Based on our observations in our formative study, most users are unwilling to adjust the position of their playback equipment once they have started their workout.

Taken together, this pre-workout analysis aims to help users prepare for the workout and configure their environment more effectively, thereby optimizing global visual access and addressing **DR1**.

4.2 Phase 2: In-Workout Multi-Device Guidance

Although *MoveMate* provides proactive pre-workout support for informed device placement, it cannot fully preempt visibility breakdowns during active exercise. This is because the initial recommendations are based on idealized instructor movements, which often deviate from users’ real-world kinematic variations. Furthermore, our placement model focuses on maximizing the human FOV for visibility, which may inadvertently compromise ergonomic comfort. Consequently, the screen may still become obstructed or awkward to view during specific postures. To bridge this gap, we propose compensatory guidance—a real-time support mechanism designed to maintain guidance continuity when visual access is compromised.

Audio-Visual Tutorial. To address the increased cognitive burden of performing complex movements without visual access to the on-screen instructions (**DR3**), we designed an AI-assisted preemptive tutorial (see Figure 7A). This design component helps users learn each movement in advance, preparing them to perform it correctly from memory during moments of limited visibility. Specifically, *MoveMate* uses AI-based video analysis to detect the temporal boundaries of each movement repetition in the original workout video. For each **dynamic** movement, it extracts a complete repetition, slows it down to 0.5× speed, and replays it five times. For each **static** movement, the system instead takes a key-frame snapshot and displays it for 10 seconds. This processed segment is then inserted prior to the corresponding movement segment in the original



Fig. 7. (A) The Audio-Visual Tutorial. (B) The Smartwatch Countdown. (C) Example images from the standard key-pose dataset for the knee push-up: (C₁) Start pose; (C₂) End pose.

video and accompanied by a dedicated AI-generated audio track that provides step-by-step guidance on how to perform the movement correctly. In addition, *MoveMate* allows users to control the tutorial from the smartwatch, where a central progress control  shows the remaining tutorial time and can be tapped to skip the entire tutorial. The draggable outer ring mirrors a video progress bar, allowing users to fast-forward or rewind within the tutorial.

Tutorial support in online workout videos is typically provided either before or during each movement. Pre-movement tutorials are often presented visually as reduced-size overlays in a corner of the screen and played at normal speed. Such designs can limit the clarity of movement demonstrations, especially on small mobile screens, making it difficult for users to follow complex limb trajectories. In-movement guidance is often delivered through verbal instructions. However, when users are unfamiliar with a movement and cannot clearly see the demonstration, audio guidance alone may be insufficient for learning the movement. To address these limitations, *MoveMate*, informed by consultations with certified fitness coaches, provides full-screen, slow-motion tutorials with dedicated audio guidance before each movement, offering a clearer and more easily perceivable learning experience. In addition, recognizing that movement difficulty is highly subjective and tied to individual skill levels, we further empower users with smartwatch-based playback agency. Instead of imposing a “one-size-fits-all” tutorial duration, *MoveMate* provides a shortcut interface on the watch. This allows experienced users to bypass redundant instructions or granularly adjust playback speed with a single tap. By shifting control from a distal playback device to the wrist, users no longer need to approach distant playback devices to make fine adjustments. By simply using the wearable smartwatch on their wrist, the process becomes significantly more streamlined.

Verbal Beat. To support users in synchronizing their movements with the instructor during moments when the on-screen instructor is not visible (DR2), we designed a real-time auditory rhythm guidance. This component delivers a series of synchronized verbal cues (e.g., “one,” “two,” “push,” “down”) to guide movement timing, enabling users to accurately follow along without needing to constantly watch the screen. Verbal Beat builds on existing practices in some online workout videos, where instructors use vocal cues to guide users’ rhythm. However, these cues were irregular in timing and sparse, which limited their usefulness as a consistent temporal reference for users. In contrast, *MoveMate* embeds rhythm cues for every repetition of all dynamic movements, ensuring continuous and consistent auditory guidance.

Smartwatch Countdown. To address the anxiety users may experience during static movements when they cannot readily check their progress on the main screen (DR4), we introduced a countdown timer on the smartwatch (see Figure 7B). This design mirrors the countdown shown on the primary display on a wrist-worn screen, providing a complementary visual channel. It targets face-down or prone movements, which account for approximately one-third of typical workout routines (see Table 3). In these movements, users’ gaze naturally shifts downward toward their hands or forearms, making the wrist-mounted device a convenient location for time-based feedback.

For example, during a push-up hold or plank, checking an on-screen timer would require lifting the head or rotating the neck—often disrupting form—whereas the smartwatch remains glanceable near the supporting hands. By reducing unnecessary head and neck movements, this design helps users maintain postural alignment and movement consistency, and may reduce strain caused by frequent screen-checking.

An existing approach to solve **DR4** is to provide a short auditory countdown (e.g., three seconds) at the end of each movement; however, this brief cue is often insufficient, leaving users without clear time awareness for extended periods during the workout. We therefore designed our own solution, and began by exploring alternatives in non-visual modalities. First, we considered auditory guidance, but the auditory channel was already allocated to the Verbal Beat. Introducing time-related audio cues would have led to modality competition and increased the risk of cognitive overload [26]. Next, we assessed the feasibility of haptic guidance. However, although advanced haptic actuators are capable of modulating intensity and frequency, commercially available smartwatches are typically limited to adjusting only pulse duration and sequence [6, 25, 28, 40]. This limitation makes it difficult for haptic cues to efficiently convey precise numerical values—such as “11 seconds remaining”—as doing so would require mapping each second (e.g., at least 30 distinct values) to a unique combination of pulse duration and sequence. Such fine-grained differentiation places excessive demands on users’ cognitive resources, making it challenging to reliably distinguish among a large set of haptic patterns, and is therefore impractical for use in real-world workout scenarios. Consequently, we returned to the visual modality and leveraged the smartwatch display as a complementary visual source to the main screen. This approach capitalizes on the visual channel’s strength in delivering precise information, while avoiding interference with ongoing auditory guidance.

4.3 System Walk-Through

To use *MoveMate*, the user first needs to upload a workout video. They can then review the changes in instructor’s face orientation and eye height through the line and bar charts (Figure 5), gaining an overview of how visibility demands vary across the workout. Using *MoveMate*’s device placement recommendation, the user can position the playback device before starting the session (Figure 6). After setting up the device and wearing the smartwatch, they can begin the workout. During the workout, the user can receive short audio-visual tutorials before each movement, follow rhythmic audio cues from the tablet, and check movement countdowns on the smartwatch.

4.4 Implementation

MoveMate was implemented as an Android-based multi-device system consisting of two client applications: a tablet application and a smartwatch application. The tablet application serves as the main interface and consists of four pages: a home page introducing the system workflow, a video upload page, a video analysis results page, and a video playback page. With the exception of the Android-native video playback page, these pages were implemented as web-based interfaces using HTML, CSS, and JavaScript, and embedded into the *MoveMate* Android application using WebView. The video analysis module was implemented in Python using several libraries, including MediaPipe for pose estimation, OpenCV for frame processing, and FFmpeg for video manipulation. The smartwatch application was implemented as an Android-based companion client. After the user opens the app, it requires no further manual setup; instead, it receives synchronization signals from the tablet via Bluetooth, including countdown updates, and tutorial-control commands.

Specifically, once a user uploads a video, *MoveMate* processes and presents visualizations and multi-device information through the following steps (see Figure 8):

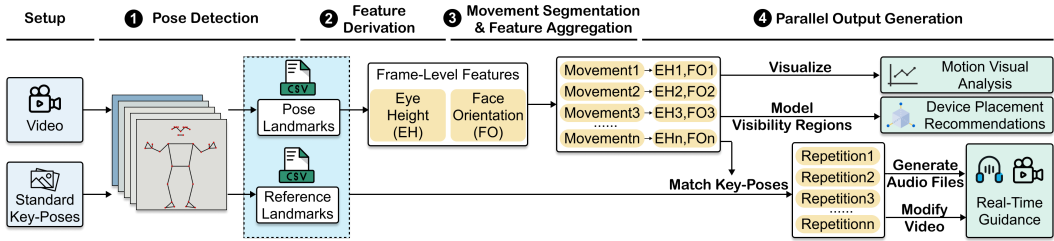


Fig. 8. Workflow of MoveMate.

1. Frame-by-Frame Pose Detection. After the user uploads a video, *MoveMate* performs pose detection using MediaPipe [22, 43]. For each frame, the system detects all candidate poses and tracks 33 anatomical keypoints per person. To ensure reliable tracking, we link poses across frames to form continuous person tracks, selecting the target individual based on temporal coverage, average confidence scores, and tracking stability. Once the target person is identified, we extract their *pose landmarks*—33 keypoints representing body joints and facial features (e.g., shoulders, elbows, hips, knees, eyes)—from each frame. These landmarks are recorded in CSV format as the system’s raw data for processing.

2. Feature Derivation. Based on the extracted *pose landmarks*, *MoveMate* computes a set of *frame-level features*, including *eye height* and *face orientation*. *Eye height* is computed from the vertical positions of the eye keypoints and normalized to a 0–1 scale. This normalization ensures comparability across users and mitigates variations caused by recording conditions such as camera angle or subject distance. *Face orientation* is estimated using geometric heuristics based on the relative positions of key facial landmarks, including the eyes, nose, and ears.

3. Movement Segmentation & Feature Aggregation. The raw *frame-level features* are first preprocessed through filtering, smoothing, interpolation, and noise reduction to remove non-exercise segments and handle missing values. We then segment the video by grouping consecutive frames with similar feature values, yielding a set of contiguous clips, each corresponding to a distinct movement. For each movement clip, we aggregate the frame-level features to obtain movement-level feature representations. Some movements exhibit relatively stable feature trajectories, whereas others show pronounced oscillations over time. So we apply Auto-Correlation Function (ACF) analysis [4] to the frame-level feature time series to capture within-movement features oscillations. For example, in Figure 5, the first **Neutral** movement and the second **Down** movement show an oscillatory pattern in eye height, and the other movements’ eye height are relatively stable.

4. Parallel Output Generation. Given the movement-level temporal boundaries and aggregated features, the system executes three parallel processing pipelines to generate three distinct outputs:

Output 1: Motion Visual Analysis. Given the aggregated *eye height* and *face orientation* signals, this pipeline generates a time-series visualization that shows how these features change throughout the workout, along with a summary of the total time spent in each face orientation category (see Figure 5).

Output 2: Device Placement Recommendations. This pipeline models the user’s visibility regions based on the aggregated values of *eye height* and *face orientation* for each movement segment. To ensure consistent and reliable analysis, the system assumes that the user’s head remains near a consistent short edge of the mat and maintains a horizontal viewing distance of approximately one arm’s length—balancing operational safety and visual clarity. For visibility modeling, the system defines horizontal visibility sectors (200°) for each detected face orientation—**Neutral**, **Up**, and **Down**—and vertical visibility sectors

(130°) based on the detected eye height. By computing the geometric intersection of these visibility regions across all movement segments, *MoveMate* identifies a unified placement region that accommodates the entire workout sequence. The final recommendation is rendered as an interactive 3D visualization using Blender (Figure 6).

Output 3: Real-Time Audio/Visual Guidance. This pipeline uses the movement-level time boundaries to constrain key-pose matching. Specifically, it applies the same normalization procedure to both the *pose landmarks* extracted from the video frames and the *reference landmarks*, and then computes a similarity score between the resulting normalized landmarks.

To build the *reference landmarks* database, we collaborated with a certified fitness instructor to capture standard key-poses for each target movement. For each **dynamic** movement, we collected images of both the start and end poses, whereas for each **static** movement, we collected images of a single representative pose. For each pose, we captured 30 images from multiple viewing angles to improve viewpoint robustness (see Figure 7C). We then extracted the reference pose landmarks using MediaPipe, applied the normalization procedure, and stored the resulting data in CSV files. Currently, the database covers movements commonly found in publicly available workout videos as well as those included in our experimental study. Each movement is assigned a unique label to support downstream voice-guidance generation.

After key-pose matching, the system performs repetition-level segmentation for **dynamic** movements by tracking the similarity between each frame's pose and the reference start/end poses. At the beginning of a repetition, similarity to the reference start pose is high; it decreases as the body moves away and increases again when the motion returns—forming a characteristic peak–trough–peak pattern over time. Conversely, similarity to the reference end pose follows the complementary pattern and peaks when the end pose is reached. By detecting these alternating local extrema and their temporal ordering, the system both selects the most likely movement label (via the start–end pair whose score trajectories best match the expected pattern) and extracts the frame indices/timestamps that define each repetition (i.e., one repetition corresponds to a full peak–trough–peak cycle). These repetition-level timestamps then drive the timing of the Verbal Beat and the Smartwatch Countdown. For **static** movements, the engine tracks how long the single reference pose is continuously matched and records the resulting hold timeframe, which is used to trigger the Smartwatch Countdown.

For the Audio-Visual Tutorial, we use the recognized movement label to prompt iFLYTEK's Spark LLM [31] to generate a short, coach-like instruction with a consistent tone:

“Briefly and clearly explain how to correctly perform a [movement label], using a coach-like tone. Begin the explanation with the sentence: ‘The next movement is [movement label].’ Keep the total response under 30 words. Do not include any conversational or unnecessary content outside of the instruction.”

We then convert the generated text into audio files using iFLYTEK's SMART-TTS [32] for real-time playback during workouts. Finally, the engine uses the repetition timestamps to crop the corresponding movement segment from the source video; we process this clip with OpenCV [5] to adjust its playback speed and insert it immediately before the original movement in the timeline, thereby creating the Audio-Visual Tutorial.

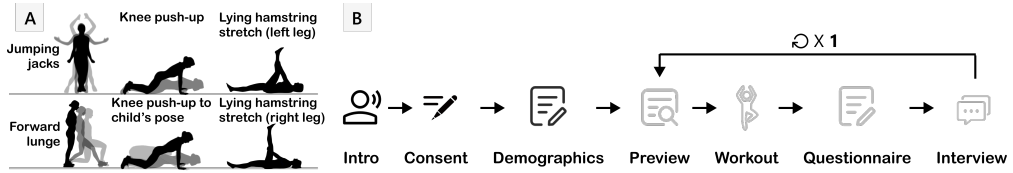


Fig. 9. (A) Movements extracted from one of the workout videos used in the experiment. From left to right, and from the first row to the second row, the movements follow their original order of appearance in the video. (B) Experimental procedure. Black steps were constant across participants. Steps highlighted in gray were counterbalanced: the order of the two conditions (*baseline* and *MoveMate*) and the assignment of the two workout videos to conditions.

5 User Study

To evaluate whether *MoveMate* addresses the four key user requirements (**DR1–DR4**) identified in our formative study, as well as the value of its design components and overall usability, we conducted a controlled experiment.

5.1 Experiment Design

We employed a within-subjects design, in which each participant completed 2 different workouts under two conditions:

- **Baseline:** Participants followed the original, unaugmented workout video. Prior to starting, they could manually navigate the video to preview the video, simulating a standard streaming workout experience.
- **MoveMate:** Participants used the full *MoveMate* system, including both the pre-workout analysis visualizations and the in-workout multi-device guidance.

To create controlled yet realistic stimuli, we collaborated with a certified fitness expert to compile two workout videos. The expert was tasked with selecting appropriate movements from publicly available YouTube workout content, ensuring that each chosen movement had a comparable counterpart in terms of physical intensity and execution complexity, forming matched pairs across the two videos. Each workout video contained six movements (see Figure 9A), with two movements each for **Up**, **Down**, and **Neutral** face orientations. This pairing strategy was implemented to control for differences between the two workout videos, ensuring that observed effects in the study were not confounded by variations in movement difficulty or execution demands. To further ensure fairness and control for cognitive load associated with visual changes [34, 50], we systematically controlled the number of visual transitions. We defined an *eye height transition* as a change greater than 0.3 (normalized) in eye height and a *face orientation transition* as a change between states (e.g., **Up** to **Down**). Both videos were constructed to contain exactly three eye height transitions and five face orientation transitions (see Figure 9A). To validate the ecological validity of this design, we analyzed 20 real-world workout videos and confirmed that the transition counts in our stimuli fell within the observed range, supporting their representativeness.

5.2 Participants

We recruited 24 participants (P1–P24; 14 women, 10 men; $M_{\text{age}} = 20.96$, $SD_{\text{age}} = 2.71$) from the local university via campus postings and social media. A screening questionnaire ensured all participants had prior experience with video-guided workouts. However, their familiarity with such workouts varied. Fourteen participants engaged in video-guided workouts at least once per week, and five of them had maintained this habit consistently for over a year. Eight participants had only tried video-guided workouts a few times and did not follow a regular workout routine.

Two participants reported engaging in such workouts approximately once per month. In terms of playback devices, 20 participants had previously used smartphones, 18 used tablets, 14 used laptops, six used televisions, and four used projectors. All participants were compensated for their time.

5.3 Procedures

The formal evaluation was conducted in a controlled environment—the university’s dance studio, consistent with our formative study. The *MoveMate* system was deployed on a Vivo Pad Air tablet (Android OS) synchronized with an OPPO Watch SE for the Smartwatch Countdown and Audio-Visual Tutorial modules. Prior to the main study, a pilot test with three co-authors was conducted to verify the technical robustness and cross-device synchronization.

The study followed a within-subjects design, lasting approximately 30 minutes per participant. The experimental workflow (Figure 9B) comprised the following stages:

1. Orientation and Consent: Upon arrival, a facilitator provided a standardized introduction using a pre-approved script to ensure instructional consistency across all sessions. Participants then provided IRB-approved informed consent and completed a demographic questionnaire regarding their background and workout habits.

2. Experimental Tasks: Participants performed the workout under two conditions: baseline and *MoveMate*. We employed full counterbalancing for both the condition order and the assignment of the two workout videos to mitigate order and learning effects. A mandatory rest period was enforced between conditions to minimize physical fatigue. Before each workout, participants previewed the upcoming routine.

- In the baseline condition, participants manually view through the video to familiarize themselves with the movements.
- In the *MoveMate* condition, they leveraged the Motion Visual Analysis and Device Placement Recommendations for strategic preparation.

Following the preview, participants independently positioned the tablet and yoga mat. To preserve ecological validity, they were allowed to select their preferred wrist for the smartwatch and adjust the audio volume. Once the workout commenced, the spatial configuration of the mat and device remained fixed to simulate real-world constraints. In addition, participants were instructed to use a Think-Aloud protocol [19] to verbalize their thoughts and feelings during the workout. Our two study facilitators observed and recorded each session, documenting participants’ tablet placement as well as any verbalized thoughts.

3. Data Collection: Each condition concluded with a post-condition questionnaire. After completing both workouts, we conducted a semi-structured interview to elicit qualitative insights into their perceived workload, flow, and the efficacy of the multi-device guidance.

5.4 Data Collection and Analysis

To evaluate how well *MoveMate* addressed the four DRs and to assess its usability, we collected both quantitative and qualitative data in the user study.

Questionnaire. The questionnaire comprised three main parts, and all rating items used a 5-point Likert scale (see Figure 10). The design of the questionnaire was informed by prior related work [13, 27, 41, 42, 44, 66]. First, participants completed the same assessment after both the baseline and *MoveMate* conditions, allowing us to examine how well *MoveMate* addressed the four DRs. The assessment covered tablet placement satisfaction (**DR1**), rhythmic synchronization (**DR2**), movement comprehension (**DR3**), and temporal awareness (**DR4**). Second, after the *MoveMate* condition, participants evaluated four core design components in terms of **comprehensibility** and **usefulness**, reflecting how easily they understood each component and how beneficial they

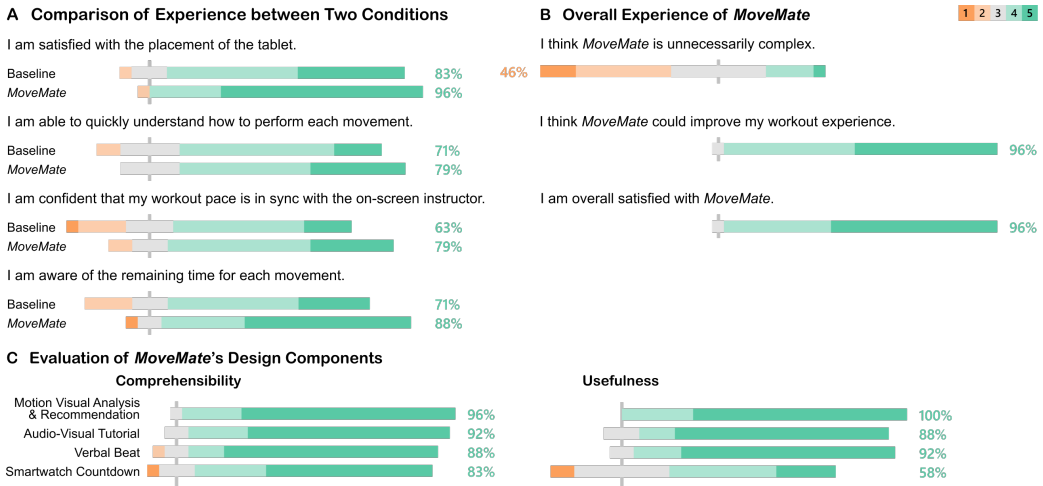


Fig. 10. Quantitative questionnaire results. Responses were collected on 5-point Likert scales (1 = Strongly Disagree, 5 = Strongly Agree). The percentages represent the proportion of positive feedback (either ≥ 4 or ≤ 2 , depending on the nature of the question).

perceived it to be. These components included Motion Visual Analysis & Recommendations, Audio-Visual Tutorial, Verbal Beat, and Smartwatch Countdown. Third, after the *MoveMate* condition, participants provided overall evaluations of *MoveMate*, including its **ease of use**, **usefulness**, and **satisfaction**. Because the questionnaire items were designed to capture study-specific perceptions, we analyzed the ratings descriptively. Specifically, we computed the proportion of positive ratings (i.e., ratings above the neutral midpoint) for each item, along with mean scores, to summarize participants' perceptions.

Interview. We conducted semi-structured interviews based on participants' questionnaire responses and the observational data collected during the two workout sessions, including their device placements and think-aloud comments. This allowed us to probe the reasons behind participants' ratings and better understand how they perceived specific design components, as well as when *MoveMate* was helpful or challenging. All interviews were audio-recorded and transcribed for subsequent analysis. Our analysis of the interview data was organized around the questionnaire structure. We used each questionnaire item as an initial topic to group relevant interview responses, and then conducted open coding within each topic to identify recurring patterns, shared explanations, and emergent subthemes. Two co-authors first reviewed the transcripts independently, then compared and refined their interpretations through discussion until they reached agreement on the final subthemes.

6 Results

Figure 10 presents the quantitative results of the user study. Combining these quantitative results with qualitative insights from the interview responses, we summarize and discuss the following findings.

6.1 Perceived Effectiveness of MoveMate in Addressing User Needs

As shown in Figure 10A, *MoveMate* received more positive ratings than the baseline across all four measures. Specifically, participants reported higher satisfaction with tablet placement (**DR1**) in the *MoveMate* condition ($M = 4.63$ vs. 4.17). This pattern was echoed in participants' interview

responses: 23 out of 24 participants described the recommended placement as appropriate, while one participant preferred the flexibility to adjust the device position during the workout. Movement comprehension (**DR3**) also improved, with participants reporting greater confidence in understanding how to perform each movement ($M = 4.13$ vs. 3.79). When explaining this improvement, participants commonly described the tutorials provided by *MoveMate* as clearer and easier to follow than standard tutorials, noting that they helped them grasp the movements more quickly. Rhythmic synchronization (**DR2**) similarly improved, with participants feeling more confident in pacing their movements in sync with the instructor when using *MoveMate* ($M = 4.00$ vs. 3.54). Participants frequently attributed this experience to the Verbal Beat, which helped them maintain pace without repeatedly looking at the screen. As P20 noted, “The rhythmic beats really helped me keep pace... I felt like I was exercising alongside the coach.” Temporal awareness (**DR4**) also improved, with participants reporting greater awareness of the remaining time ($M = 4.38$ vs. 3.79). Participants linked this improvement to the combined use of Device Placement Recommendation and Smartwatch Countdown. Many noted that the recommended device placement made the screen easier to view, allowing them to access the on-screen timer more effortlessly throughout the workout. They also emphasized that Smartwatch Countdown was particularly helpful during prone movements, as it enabled them to quickly check the remaining time with a glance at the wrist, rather than straining to turn their head toward the main screen. Together, these findings indicate that *MoveMate* effectively addressed the core user needs identified in the formative study.

6.2 Feedback on *MoveMate*’s Design Components

- Motion Visual Analysis and Device Placement Recommendation received the highest ratings among design components in both comprehensibility and usefulness, with mean scores of 4.71 and 4.75, respectively (Figure 10C). Participants emphasized that Motion Visual Analysis and Recommendations were easy to interpret and practically valuable during pre-workout preparation. For example, P10 noted, “...I trusted the algorithm more than my own judgment, it saves me the trouble of figuring things out on my own.” Likewise, P17 highlighted that the Device Placement Recommendations prompted them to reflect on factors they had previously overlooked: “I didn’t really think about where to put my device until I saw that interface. Then I tried the recommended spot, and honestly, it felt way better than how I usually set it up. I just wish I’d seen that earlier.” Despite these strengths, initial use demanded a longer preview time ($M = 5.5min$) compared to the baseline ($M = 1.5min$). Some participants acknowledged this learning curve but expected improvement with familiarity. As P7 remarked, “Since it was my first time, I was checking out everything super carefully. But next time, I think it’ll be way faster. Now that I’ve used it once, I kinda know what to pay attention to.”
- Audio-Visual Tutorial received high ratings in both comprehensibility and usefulness, with mean scores of 4.63 for each (Figure 10C). Most participants (21/24) endorsed the positive role of the pre-movement tutorial. They noted that it not only offered a moment to rest and recover, but also helped them understand how to perform the next movement. However, three participants questioned the value of the spoken explanations, especially for simple movements. As P11 remarked, “The exercises were super simple, so just seeing the demonstration was enough. The voice-overs didn’t really help much for me.” Similarly, P19 pointed out that for static stretching movements, what they needed was not instructions on how to perform the movement, but rather guidance on muscle engagement and the sensations they should feel.
- Verbal Beat also received high ratings in both comprehensibility and usefulness, with mean scores of 4.58 and 4.67, respectively (Figure 10C). Most participants (22/24)—particularly those with regular experience in video-guided home workouts—reported benefits. They described that the rhythmic verbal prompts reduced their reliance on the screen, allowing them to stay more

immersed in their movement—focusing on breathing, form, and bodily awareness rather than visual cues. However, P9 and P10 noted that the fixed pacing could introduce pressure during physically demanding movements. For example, P9 shared: “*I actually found the verbal counts super helpful... I didn’t have to keep staring at the screen,*” but added, “*When I did push-up to child’s pose, I couldn’t keep up with the coach’s pace. The ‘one, two’ counts kept rushing me... I wanted to go at my own speed instead.*” These reflections suggest that Verbal Beat can effectively reduce screen reliance, but rigid timing may hinder autonomy during more challenging movements.

- Smartwatch Countdown received a high mean score of 4.33 for comprehensibility but a relatively lower mean score of 3.63 for usefulness (Figure 10C). Five participants questioned the usefulness of the Smartwatch Countdown, expressing uncertainty about its purpose or added value. They noted that since they needed to watch the tablet to follow the movements, splitting attention between two devices often felt burdensome. As a result, they tended to only rely on the tablet—even if it meant twisting their posture to keep it in view. As P14, who had less experience with video-guided workouts, remarked: “*To be honest, I never looked at the watch. The screen was just easier—it showed the timing and what the movement looked like. That’s what I really needed.*” In contrast, five experienced participants praised the Smartwatch Countdown, particularly during prone movements. In prone postures, the screen was often out of sight and difficult to access, while the wrist—and therefore the smartwatch—remained naturally visible. Since prone movements tend to be physically demanding, the Smartwatch Countdown allowed participants to stay in position and focus on execution without expending extra effort to check the screen for timing. As P5 explained, “*...it gave me a sense of reassurance, like I didn’t have to worry about timing—I could just focus all my energy on doing the movement right.*”

6.3 Perceived Overall Experience with MoveMate

Overall, *MoveMate* was positively received, with 23 out of 24 participants giving positive ratings for both usefulness ($M = 4.46$) and satisfaction ($M = 4.54$). This positive trend was consistently reflected in participants’ comments, with many highlighting the system’s benefits and expressing a willingness to use it in the future. Perceptions of system complexity were more mixed. Only 46% of participants perceived *MoveMate* as uncomplicated. When elaborating on this, participants pointed to multiple contributing factors. Some noted that the Motion Visual Analysis and Device Placement Recommendation interfaces required effort to learn. At the same time, these participants often added that this initial learning burden would likely decrease with familiarity and repeated use. Some associated the perceived complexity with the distributed setup itself, explaining that interacting with both a tablet and a smartwatch felt more demanding than relying on a single device. Despite this, several participants considered this trade-off worthwhile given the additional support provided by the system. Participants also expressed a desire for greater customization. As P20 explained, “*The pacing cues were the most useful part for me. If I could just keep that and turn off the rest, that’d be ideal. The other stuff didn’t really do much for me—I guess it’s more for beginners.*”

7 Discussion

Based on results in section 6, we found that *MoveMate* effectively addressed the four identified **DRs**. In addition, participants generally perceived the system’s design components as both comprehensible and useful, and they also reported a high level of overall satisfaction with the system. These findings, together with our observations and reflections from the design and implementation of the system, form the basis for our subsequent discussion.

7.1 Reflections on Computer-Assisted Physical Training Systems

7.1.1 Supporting the Pre-workout Phase through Proactive Visualization. While existing research in computer-assisted training has focused extensively on providing feedback either during exercise (real-time) [13, 66] or after completion (reflective summaries) [20, 47], our study identifies a critical design opportunity in the pre-workout phase. Through *MoveMate*, we explored how proactive visualizations—such as summaries of an instructor’s spatial dynamics and device placement guides—can support users in their physical and mental preparation. Our findings from interviews and questionnaires indicate that this “before-hand” information is highly valued, particularly in home-based settings. Users often face a trial-and-error process when selecting workouts, which can lead to poor experiences or even potential injuries if the content does not match their physical ability or available space. By providing a visual abstract on everyday devices, *MoveMate* allows users to quickly screen and select the most suitable exercises.

Looking forward, we suggest that fitness platforms could integrate automated motion extraction to generate motion thumbnails. Rather than relying on static images, these thumbnails could serve as interactive previews that help users assess the space requirements and physical demands of a video at a glance. This shift toward proactive visualization makes the discovery process more efficient and safer for self-directed home training.

7.1.2 Personalized and Context-Aware Tutorial Support. Our formative study revealed a critical pain point **DR2** in home-based training: when performing complex movements that involve visual occlusion, users frequently experience heightened stress and frustration. This suggests that guidance should not be constant, but rather judiciously triggered during these high-demand moments. Currently, *MoveMate* addresses this by providing supplementary visual and audio guidance and tutorials for every movement. While effective for initial learning, this “one-size-fits-all” approach can lead to interaction redundancy, where experienced users find the persistent assistance unnecessary or even intrusive. Even though we have provided wearable devices as shortcuts for skipping or adjusting the speed for tutorial, they still impose an additional time burden on certain actions for both expert and general users. To move toward a more intelligent and fluid system, we propose two optimization strategies:

- **Historical Data-Driven Adaptation:** For long-term use, the system could dynamically adjust the level of support by analyzing a user’s proficiency profile. If historical data indicates that a user has mastered a specific movement, the system would suppress extra guidance to maintain a streamlined interface. Conversely, for novel movements—especially those identified as visually restrictive—the system would proactively overlay tutorial prompts to scaffold the learning process.
- **Context-Aware Contingency Support:** Context-aware support could be implemented by using the tablet’s front-facing camera to track the user’s movement state. Our device placement recommendation was based on the broader human FoV; by comparison, commodity tablet cameras generally have narrower FoVs [1, 29, 53, 69]. This difference suggests that when the tablet camera loses track of the user, it may also coincide with moments in which the screen falls outside the user’s view or remains visible at the edge of their visual range, making it difficult to view comfortably. The system could therefore use such tracking failures as contextual cues to automatically pivot to rhythmic audio cues or tutorials.

Ultimately, such an evolution transforms the system from a static instructional tool into a dynamic, responsive partner that provides support precisely when and where the user’s physical and environmental context demands it.

7.1.3 Multi-Device Systems. *MoveMate* employs a multi-device configuration to distribute interaction between a stationary tablet and a body-worn smartwatch. While this setup offers flexibility, the Smartwatch Countdown received the lowest usefulness rating ($M = 3.63$). One possible explanation is that the countdown was presented across all movements, although its practical value was most evident during prone movements, where the tablet screen often entered a visual blind spot while the smartwatch remained naturally visible through a quick wrist glance. This pose-dependent usefulness may have fallen short of some participants' expectations for consistent support throughout the workout, contributing to its relatively lower rating. This suggests that in sports interaction, the utility of a device is not static but fluctuates with the user's physical orientation and proprioceptive constraints. Rather than a generic coordination issue, the challenge lies in the dynamic handoff of interaction cues. Future multi-device fitness systems should move beyond a persistent dual-display model. Instead, they should implement context-aware redundancy management: automatically suppressing wearable feedback when the primary display is visible, and escalating it to the wrist only when visual occlusion is detected. This approach would reduce attention fragmentation [72] while ensuring that critical guidance remains accessible during the most physically demanding segments of a workout.

7.2 Generalizability of the Design Paradigm

The architectural significance of *MoveMate* extends beyond the specific tablet–smartwatch implementation; it represents a generalized framework for cross-device fitness support. At its core, this paradigm leverages the synergy between a fixed playback device (e.g., TV, projector, or smartphone) and a proximal wearable to ensure informational continuity. The fundamental motivation for this framework is the inevitable visual occlusion inherent in home-based exercise. Regardless of the screen size or form factor, no static display can guarantee constant visibility across the full spectrum of human movement—such as prone, supine, or lateral poses. In these “blind spots,” the primary display fails as an effective information channel. By formalizing the relationship between these two device classes, *MoveMate* provides a scalable solution to this problem:

- **Decoupled Feedback:** The framework treats the wearable not as a redundant screen, but as a complementary display that fills the gaps when the user's gaze is forced away from the primary display.
- **Flexibility Across Device Setups:** This model is easily portable to any ubiquitous device ecosystem. Whether paired with a large-scale living room TV or a mobile smartphone, the underlying logic—using wearables to provide countdown and rhythmic cues—remains a robust strategy for mitigating the frustrations of visual disconnection.

Ultimately, *MoveMate* offers a design direction for future streaming and fitness platforms to transition from single-device video delivery to a distributed, multi-device guidance ecosystem that adapts to the physical constraints of the human body in motion.

7.3 Limitations

Despite having reflected extensively throughout the design, development, and user study phases of our system, we acknowledge that our research still contains a number of limitations. First, most of our participants (22/24) were young people. This sample was appropriate for our study because young adults are a common audience for video-guided workouts. However, university students may also be more accustomed to using digital interfaces and interpreting data-driven visual representations. If *MoveMate* is extended to broader populations, such as older adults, these users may encounter greater difficulty in understanding visual elements such as line charts, as they may be less accustomed to reading analytical visuals in everyday contexts. Second, *MoveMate* requires

standard key poses for each movement to support reliable detection and segmentation. The current implementation supports only a limited set of movements due to the limited set of trained dataset. While this was sufficient for ensuring reliable system performance in our evaluation, broader real-world deployment would require a substantially larger and more diverse movement dataset. Third, our evaluation focused primarily on subjective perceptions and qualitative experiences, which provided a direct way to examine whether *MoveMate* addressed the user needs identified in our formative study and improved users' workout experience. However, future work could complement these findings with additional objective metrics, such as rhythm synchronization accuracy, interruption frequency, and the frequency of compensatory neck or torso movements. Such measures could offer a more fine-grained account of users' behavioral responses while interacting with *MoveMate* during workouts.

8 Conclusion

Video-guided physical training is inherently constrained by the fixed orientation of playback screens, which frequently leads to a breakdown in instructional continuity as users transition through diverse body postures. In this work, we introduced *MoveMate*, a system designed to bridge this gap through a dual-phase interaction paradigm. First, *MoveMate* facilitates proactive spatial preparation by analyzing instructor dynamics—such as eye height and face orientation—to provide context-aware Device Placement Recommendations. Second, it maintains support during active exercise via a distributed multi-device interface, combining preemptive Audio-Visual Tutorials with Verbal Beat cues and proximal Smartwatch Countdowns. Our user study demonstrates that this approach significantly enhances the exercise experience by ensuring that guidance remains accessible even during off-screen postures and visual blind spots. By addressing the physical and environmental constraints of at-home training, *MoveMate* provides a foundation for future research into computer-assisted physical training systems.

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References

- [1] Apple Inc. 2021. iPad Pro Technical Specifications. <https://www.apple.com/ipad-pro/specs/>. Accessed: 2026-04-28.
- [2] BBC. 2023. Covid-19 has forced us to exercise indoors. But home fitness has been shaping our lives for decades. <https://www.bbc.com/worklife/article/20200504-covid-19-update-quarantine-home-workouts-during-coronavirus> Accessed: 2024-12-05.
- [3] Mark J. Berentsen, Marit Bentvelzen, and Paweł W. Woźniak. 2021. MTBalance: Assisting Novice Mountain Bikers with Real-Time Proprioceptive Feedback. *Proceedings of the Association for Computing Machinery on Human-Computer Interaction* 5, ISS, Article 506 (Nov. 2021), 25 pages. doi:10.1145/3488551
- [4] George E. P. Box, Gwilym M. Jenkins, Gregory C. Reinsel, and Greta M. Ljung. 2015. *Time Series Analysis: Forecasting and Control* (5 ed.). John Wiley & Sons, Hoboken, New Jersey.
- [5] Gary Bradski. 2000. The opencv library. *Dr. Dobb's Journal: Software Tools for the Professional Programmer* 25, 11 (2000), 120–123.
- [6] Stephen Brewster and Lorna M. Brown. 2004. Tactons: structured tactile messages for non-visual information display. In *Proceedings of the Fifth Conference on Australasian User Interface (AUI '04, Vol. 28)*. Australian Computer Society, Barangaroo, NSW, Australia, 15–23.
- [7] Alberto Cannavò, Filippo Gabriele Praticò, Giuseppe Ministeri, and Fabrizio Lamberti. 2018. A Movement Analysis System based on Immersive Virtual Reality and Wearable Technology for Sport Training. In *Proceedings of the 4th International Conference on Virtual Reality (ICVR 2018)*. Association for Computing Machinery, New York, USA, 26–31. doi:10.1145/3198910.3198917
- [8] Jessica R. Cauchard, Janette L. Cheng, Thomas Pietrzak, and James A. Landay. 2016. ActiVibe: Design and Evaluation of Vibrations for Progress Monitoring. In *Proceedings of the 2016 CHI Conference on Human Factors in Computing Systems (CHI '16)*. Association for Computing Machinery, New York, USA, 3261–3271. doi:10.1145/2858036.2858046

- [9] Daniel Cesarini, Davide Calvaresi, Chiara Farnesi, Diego Taddei, Stefano Frediani, Bodo E. Ungerechts, and Thomas Hermann. 2016. MEDIATION: An eMbEddeD System for Auditory Feedback of Hand-water InterAction while Swimming. *Procedia Engineering* 147 (2016), 324–329. doi:10.1016/j.proeng.2016.06.301
- [10] Jacky C.P. Chan, Howard Leung, Jeff K.T. Tang, and Taku Komura. 2011. A Virtual Reality Dance Training System Using Motion Capture Technology. *Institute of Electrical and Electronics Engineers Transactions on Learning Technologies* 4, 2 (2011), 187–195. doi:10.1109/TLT.2010.27
- [11] Xinyu Chen, Yuqi Li, Jintao Chen, Jiabao Li, Chong Wang, and Pinyan Tang. 2024. Enhancing Home Exercise Experiences with Video Motion-Tracking for Automatic Display Height Adjustment. In *Proceedings of the 2024 CHI Conference on Human Factors in Computing Systems (CHI '24)*. Association for Computing Machinery, New York, USA, Article 310, 13 pages. doi:10.1145/3613904.3642936
- [12] Dewen Cheng, Qiwei Wang, Yue Liu, Hailong Chen, Dongwei Ni, Ximeng Wang, Cheng Yao, Qichao Hou, Weihong Hou, Gang Luo, et al. 2021. Design and manufacture AR head-mounted displays: A review and outlook. *Light: Advanced Manufacturing* 2, 3 (2021), 350–369. doi:10.37188/lam.2021.024
- [13] Liqi Cheng, Hanze Jia, Lingyun Yu, Yihong Wu, Shuainan Ye, Dazhen Deng, Hui Zhang, Xiao Xie, and Yingcai Wu. 2024. VisCourt: In-Situ Guidance for Interactive Tactic Training in Mixed Reality. In *Proceedings of the 37th Annual Association for Computing Machinery Symposium on User Interface Software and Technology (UIST '24)*. Association for Computing Machinery, New York, USA, Article 32, 14 pages. doi:10.1145/3654777.3676466
- [14] Christopher Clarke, Doga Cavdir, Patrick Chiu, Laurent Denoue, and Don Kimber. 2020. Reactive Video: Adaptive Video Playback Based on User Motion for Supporting Physical Activity. In *Proceedings of the 33rd Annual Association for Computing Machinery Symposium on User Interface Software and Technology (UIST '20)*. Association for Computing Machinery, New York, USA, 196–208. doi:10.1145/3379337.3415591
- [15] Jacob Cohen. 1960. A Coefficient of Agreement for Nominal Scales. *Educational and Psychological Measurement* 20, 1 (1960), 37–46. doi:10.1177/001316446002000104
- [16] Gray Cook, Lee Burton, Barbara J. Hoogenboom, and Michael Voight. 2014. Functional Movement Screening: The Use of Fundamental Movements as an Assessment of Function – Part 1. *International Journal of Sports Physical Therapy* 9, 3 (2014), 396–409. <https://pmc.ncbi.nlm.nih.gov/articles/PMC4060319/>
- [17] Bidit L. Dey, Dorothy Yen, and Lalnunpuia Samuel. 2020. Digital consumer culture and digital acculturation. *International Journal of Information Management* 51 (2020), 102057. doi:10.1016/j.ijinfomgt.2019.102057
- [18] Tim Duente, Dennis Stanke, Moritz Klose, Benjamin Simon, Ibraheem Al-Azzawi, and Michael Rohs. 2024. Shock Me The Way: Directional Electrotactile Feedback under the Smartwatch as a Navigation Aid for Cyclists. *Proceedings of the Association for Computing Machinery on Human-Computer Interaction* 8, MHCI, Article 274 (Sept. 2024), 25 pages. doi:10.1145/3676521
- [19] K. Anders Ericsson. 2017. *Protocol Analysis*. John Wiley & Sons, Ltd, Hoboken, New Jersey, Chapter 33, 425–432. doi:10.1002/9781405164535.ch33
- [20] Bettina Eska, Marco Philip, Pawel W. Wozniak, Albrecht Schmidt, and Jakob Karolus. 2023. [Don't] Let The Bodies HIIT The Floor: Fostering Body Awareness in Fast-Paced Physical Activity Using Body-Worn Sensors. *Proceedings of the Association for Computing Machinery on Human-Computer Interaction* 7, MHCI, Article 203 (Sept. 2023), 27 pages. doi:10.1145/3604250
- [21] Mihai Fieraru, Mihai Zanfir, Silviu Cristian Pirlea, Vlad Olaru, and Cristian Sminchisescu. 2021. Aifit: Automatic 3d human-interpretable feedback models for fitness training. In *Proceedings of the IEEE/CVF conference on computer vision and pattern recognition*. Institute of Electrical and Electronics Engineers, Piscataway, New Jersey, USA, 9919–9928.
- [22] Google AI Edge. 2025. MediaPipe Solutions Guide. <https://ai.google.dev/edge/mediapipe/solutions/guide> Accessed: 2025-12-31.
- [23] Tobias Großhauser, Bettina Bläsing, Corinna Spieth, and Thomas Hermann. 2012. Wearable sensor-based real-time sonification of motion and foot pressure in dance teaching and training. *Journal of the Audio Engineering Society* 60, 7 (2012), 580.
- [24] Ping-Hsuan Han, Yang-Sheng Chen, Yilun Zhong, Han-Lei Wang, and Yi-Ping Hung. 2017. My Tai-Chi coaches: an augmented-learning tool for practicing Tai-Chi Chuan. In *Proceedings of the 8th Augmented Human International Conference (AH '17)*. Association for Computing Machinery, New York, USA, Article 25, 4 pages. doi:10.1145/3041164.3041194
- [25] Li He. 2023. Unveiling the touch revolution: how haptic feedback elevates wearable device experiences. doi:10.48550/arXiv.2307.16179
- [26] Xiaoyang He and Dan Wu. 2023. Does additional audio really work? A study on users' cognitive behavior with audio-visual dual-channel in panoramic digital museum. *Information & Management* 60, 4 (2023), 103791. doi:10.1016/j.im.2023.103791
- [27] Yuchen He, Jianbing Lv, Liqi Cheng, Lingyu Meng, Dazhen Deng, and Yingcai Wu. 2025. ProTAL: A Drag-and-Link Video Programming Framework for Temporal Action Localization. In *Proceedings of the 2025 CHI Conference on Human*

- Factors in Computing Systems (CHI '25)*. Association for Computing Machinery, New York, USA, Article 1140, 18 pages. doi:10.1145/3706598.3713741
- [28] Eve Hoggan, Andrew Crossan, Stephen A. Brewster, and Topi Kaaresoja. 2009. Audio or tactile feedback: which modality when?. In *Proceedings of the Association for Computing Machinery Special Interest Group on Computer-Human Interaction Conference on Human Factors in Computing Systems (CHI '09)*. Association for Computing Machinery, New York, USA, 2253–2256. doi:10.1145/1518701.1519045
- [29] Ian P. Howard and Brian J. Rogers. 1996. *Binocular Vision and Stereopsis*. Oxford University Press, Oxford, England. doi:10.1093/acprof:oso/9780195084764.001.0001
- [30] Elena Iannucci, Zhu-Tian Chen, Iro Armeni, Marc Pollefeys, Hanspeter Pfister, and Johanna Beyer. 2023. Arrow: A real-time ar rowing coach. In *EuroVis 2023-Short Papers*. Eurographics Association, Aire-la-Ville, Switzerland, 73–77. doi:10.3929/ethz-b-000643509
- [31] iFLYTEK Open Platform. 2025. iFLYTEK Spark LLM. <https://xinghuo.xfyun.cn/> Accessed: 2025-12-31.
- [32] iFLYTEK Open Platform. 2025. Super Human-like Text-to-Speech (SMART-TTS). <https://www.xfyun.cn/services/smart-tts> Accessed: 2025-12-31.
- [33] Kodai Ito, Mitsunori Tada, Hiroyasu Ujike, and Keiichiro Hyodo. 2021. Effects of the Weight and Balance of Head-Mounted Displays on Physical Load. *Applied Sciences* 11, 15, Article 6802 (2021), 12 pages. doi:10.3390/app11156802
- [34] Georg Jahn, Julia Wendt, Martin Lotze, Frank Papenmeier, and Markus Huff. 2012. Brain activation during spatial updating and attentive tracking of moving targets. *Brain and cognition* 78, 2 (2012), 105–113.
- [35] Hye-Young Jo, Laurenz Seidel, Michel Pahud, Mike Sinclair, and Andrea Bianchi. 2023. FlowAR: How Different Augmented Reality Visualizations of Online Fitness Videos Support Flow for At-Home Yoga Exercises. In *Proceedings of the 2023 CHI Conference on Human Factors in Computing Systems (CHI '23)*. Association for Computing Machinery, New York, USA, Article 469, 17 pages. doi:10.1145/3544548.3580897
- [36] Jiha Kim, Younho Nam, Jungeun Lee, Young-Joo Suh, and Inseok Hwang. 2023. ProxiFit: Proximity Magnetic Sensing Using a Single Commodity Mobile toward Holistic Weight Exercise Monitoring. *Proceedings of the Association for Computing Machinery on Interactive, Mobile, Wearable and Ubiquitous Technologies* 7, 3, Article 105 (Sept. 2023), 32 pages. doi:10.1145/3610920
- [37] Minseong Kim. 2022. How can I Be as attractive as a Fitness YouTuber in the era of COVID-19? The impact of digital attributes on flow experience, satisfaction, and behavioral intention. *Journal of Retailing and Consumer Services* 64 (2022), 102778. doi:10.1016/j.jretconser.2021.102778
- [38] Carolyn Kisner and Lynn Allen Colby. 2002. *THERAPEUTIC EXERCISE Foundations and Techniques*. 5th. Philadelphia: FA Davis Company, Philadelphia, Pennsylvania, USA.
- [39] Joseph J. LaViola. 2000. A discussion of cybersickness in virtual environments. *Association for Computing Machinery Special Interest Group on Computer-Human Interaction Bulletin* 32, 1 (Jan. 2000), 47–56. doi:10.1145/333329.333344
- [40] Sunwoo Lee, Wonsuk Choi, and Dong Hoon Lee. 2023. The vibration knows who you are! A further analysis on usable authentication for smartwatch users. *Computers & Security* 125 (2023), 103040. doi:10.1016/j.cose.2022.103040
- [41] Tica Lin, Rishi Singh, Yalong Yang, Carolina Nobre, Johanna Beyer, Maurice A. Smith, and Hanspeter Pfister. 2021. Towards an Understanding of Situated AR Visualization for Basketball Free-Throw Training. In *Proceedings of the 2021 CHI Conference on Human Factors in Computing Systems (CHI '21)*. Association for Computing Machinery, New York, USA, Article 461, 13 pages. doi:10.1145/3411764.3445649
- [42] Jingyuan Liu, Nazmus Saquib, Chen Zhutian, Rubaiat Habib Kazi, Li-Yi Wei, Hongbo Fu, and Chiew-Lan Tai. 2024. PoseCoach: A Customizable Analysis and Visualization System for Video-Based Running Coaching. *Institute of Electrical and Electronics Engineers Transactions on Visualization and Computer Graphics* 30, 7 (July 2024), 3180–3195. doi:10.1109/TVCG.2022.3230855
- [43] Camillo Lugaresi, Jiuqiang Tang, Hadon Nash, Chris McClanahan, Esha Uboweja, Michael Hays, Fan Zhang, Chuo-Ling Chang, Ming Guang Yong, Juhyun Lee, Wan-Teh Chang, Wei Hua, Manfred Georg, and Matthias Grundmann. 2019. MediaPipe: A Framework for Building Perception Pipelines. doi:10.48550/arXiv.1906.08172
- [44] Dizhi Ma, Xiyun Hu, Jingyu Shi, Mayank Patel, Rahul Jain, Ziyi Liu, Zhengzhe Zhu, and Karthik Ramani. 2024. avaTTAR: Table Tennis Stroke Training with Embodied and Detached Visualization in Augmented Reality. In *Proceedings of the 37th Annual Association for Computing Machinery Symposium on User Interface Software and Technology (UIST '24)*. Association for Computing Machinery, New York, USA, Article 35, 16 pages. doi:10.1145/3654777.3676400
- [45] Nora McDonald, Sarita Schoenebeck, and Andrea Forte. 2019. Reliability and Inter-rater Reliability in Qualitative Research: Norms and Guidelines for CSCW and HCI Practice. *Proceedings of the Association for Computing Machinery on Human-Computer Interaction* 3, CSCW, Article 72 (Nov. 2019), 23 pages. doi:10.1145/3359174
- [46] A. Nakamura, S. Tabata, T. Ueda, S. Kiyofuji, and Y. Kuno. 2005. Dance training system with active vibro-devices and a mobile image display. In *2005 Institute of Electrical and Electronics Engineers /RSJ International Conference on Intelligent Robots and Systems*. Institute of Electrical and Electronics Engineers, Piscataway, New Jersey, USA, 3075–3080. doi:10.1109/IROS.2005.1545565

- [47] Christopher Napier, Christopher L MacLean, Jessica Maurer, Jack E Taunton, and Michael A Hunt. 2019. Real-time biofeedback of performance to reduce braking forces associated with running-related injury: An exploratory study. *Journal of orthopaedic & sports physical therapy* 49, 3 (2019), 136–144.
- [48] Joseph W. Newbold, Nadia Bianchi-Berthouze, Nicolas E. Gold, Ana Tajadura-Jiménez, and Amanda CdC Williams. 2016. Musically Informed Sonification for Chronic Pain Rehabilitation: Facilitating Progress & Avoiding Over-Doing. In *Proceedings of the 2016 CHI Conference on Human Factors in Computing Systems (CHI '16)*. Association for Computing Machinery, New York, USA, 5698–5703. doi:10.1145/2858036.2858302
- [49] Quang-Tien Pham, Duc-Anh Nguyen, Tien-Thanh Nguyen, Thanh Nam Nguyen, Duy-Tung Nguyen, Dinh-Tan Pham, Thanh Hai Tran, Thi-Lan Le, and Hai Vu. 2022. A study on skeleton-based action recognition and its application to physical exercise recognition. In *Proceedings of the 11th International Symposium on Information and Communication Technology (SoICT '22)*. Association for Computing Machinery, New York, USA, 239–246. doi:10.1145/3568562.3568639
- [50] Clark C Presson and Daniel R Montello. 1994. Updating after rotational and translational body movements: Coordinate structure of perspective space. *Perception* 23, 12 (1994), 1447–1455.
- [51] Fazlay Rabbi, Taiwoo Park, Biyi Fang, Mi Zhang, and Youngki Lee. 2018. When Virtual Reality Meets Internet of Things in the Gym: Enabling Immersive Interactive Machine Exercises. *Proceedings of the Association for Computing Machinery on Interactive, Mobile, Wearable and Ubiquitous Technologies* 2, 2, Article 78 (July 2018), 21 pages. doi:10.1145/3214281
- [52] Michael A. Rupp. 2024. Is it getting hot in here? The effects of VR headset microclimate temperature on perceived thermal discomfort, VR sickness, and skin temperature. *Applied Ergonomics* 114 (Jan 2024), 104128. doi:10.1016/j.apergo.2023.104128
- [53] Samsung Electronics. 2022. Galaxy Tab S8 Specifications. <https://deviceguides.ee.co.uk/samsung/galaxy-tab-s8-android-12-0/specifications/>. Accessed: 2026-04-28.
- [54] Nina Schaffert and Klaus Mattes. 2015. Interactive Sonification in Rowing: Acoustic Feedback for On-Water Training. *Institute of Electrical and Electronics Engineers MultiMedia* 22, 1 (2015), 58–67. doi:10.1109/MMUL.2015.9
- [55] Luca Scrucca and Dimitris Karlis. 2025. A model-based approach to shot charts estimation in basketball. *Computational Statistics* 40, 4 (2025), 2031–2048.
- [56] Alessandra Semeraro and Laia Turmo Vidal. 2022. Visualizing Instructions for Physical Training: Exploring Visual Cues to Support Movement Learning from Instructional Videos. In *Proceedings of the 2022 CHI Conference on Human Factors in Computing Systems (CHI '22)*. Association for Computing Machinery, New York, USA, Article 90, 16 pages. doi:10.1145/3491102.3517735
- [57] Ana Tajadura-Jimenez, Judith Ley-Flores, Omar Valdiviezo, Aneasha Singh, Milagrosa Sanchez-Martin, Joaquin Diaz Duran, and Elena Márquez Segura. 2022. Exploring the Design Space for Body Transformation Wearables to Support Physical Activity through Sensitizing and Bodystorming. In *Proceedings of the 8th International Conference on Movement and Computing (MOCO '22)*. Association for Computing Machinery, New York, USA, Article 23, 9 pages. doi:10.1145/3537972.3538001
- [58] Richard Tang, Xing-Dong Yang, Scott Bateman, Joaquim Jorge, and Anthony Tang. 2015. Physio@Home: Exploring Visual Guidance and Feedback Techniques for Physiotherapy Exercises. In *Proceedings of the 33rd Annual Association for Computing Machinery Conference on Human Factors in Computing Systems (CHI '15)*. Association for Computing Machinery, New York, USA, 4123–4132. doi:10.1145/2702123.2702401
- [59] Atima Tharatipyakul, Kenny T. W. Choo, and Simon T. Perrault. 2020. Pose Estimation for Facilitating Movement Learning from Online Videos. In *Proceedings of the International Conference on Advanced Visual Interfaces (AVI '20)*. Association for Computing Machinery, New York, USA, Article 64, 5 pages. doi:10.1145/3399715.3399835
- [60] Laia Turmo Vidal, Elena Márquez Segura, Luis Parrilla Bel, and Annika Waern. 2020. Training Technology Probes Across Fitness Practices: Yoga, Circus and Weightlifting. In *Extended Abstracts of the 2020 CHI Conference on Human Factors in Computing Systems (CHI EA '20)*. Association for Computing Machinery, New York, USA, 1–8. doi:10.1145/3334480.3382862
- [61] Laia Turmo Vidal, Ana Tajadura-Jiménez, José Manuel Vega-Cebrián, Judith Ley-Flores, Joaquín R. Díaz-Durán, and Elena Márquez Segura. 2024. Body Transformation: An Experiential Quality of Sensory Feedback Wearables for Altering Body Perception. In *Proceedings of the Eighteenth International Conference on Tangible, Embedded, and Embodied Interaction (TEI '24)*. Association for Computing Machinery, New York, USA, Article 25, 19 pages. doi:10.1145/3623509.3633373
- [62] vivo. 2026. vivo Pad Air. <https://www.vivo.com.cn/vivo/vivopadair> Accessed: 2025-12-29.
- [63] Hanchen David Wang and Meiyi Ma. 2023. PhysiQ: Off-site Quality Assessment of Exercise in Physical Therapy. *Proceedings of the Association for Computing Machinery on Interactive, Mobile, Wearable and Ubiquitous Technologies* 6, 4, Article 208 (Jan. 2023), 25 pages. doi:10.1145/3570349
- [64] WashingtonPost. 2023. The pandemic's home-workout revolution may be here to stay. <https://www.washingtonpost.com/road-to-recovery/2021/01/07/home-fitness-boom/> Accessed: 2024-12-05.

- [65] Mikołaj P. Woźniak, Julia Dominiak, Michał Pieprzowski, Piotr Ładoński, Krzysztof Grudzień, Lars Lischke, Andrzej Romanowski, and Paweł W. Woźniak. 2020. Subtletee: Augmenting Posture Awareness for Beginner Golfers. *Proceedings of the Association for Computing Machinery on Human-Computer Interaction* 4, ISS, Article 204 (Nov. 2020), 24 pages. doi:10.1145/3427332
- [66] Yihong Wu, Lingyun Yu, Jie Xu, Dazhen Deng, Jiachen Wang, Xiao Xie, Hui Zhang, and Yingcai Wu. 2023. AR-Enhanced Workouts: Exploring Visual Cues for At-Home Workout Videos in AR Environment. In *Proceedings of the 36th Annual Association for Computing Machinery Symposium on User Interface Software and Technology (UIST '23)*. Association for Computing Machinery, New York, USA, Article 121, 15 pages. doi:10.1145/3586183.3606796
- [67] Chengshuo Xia, Xinrui Fang, Riku Arakawa, and Yuta Sugiura. 2022. VoLearn: A Cross-Modal Operable Motion-Learning System Combined with Virtual Avatar and Auditory Feedback. *Proceedings of the Association for Computing Machinery on Interactive, Mobile, Wearable and Ubiquitous Technologies* 6, 2, Article 81 (jul 2022), 26 pages. doi:10.1145/3534576
- [68] Chengshuo Xia, Tian Min, and Yuta Sugiura. 2024. AudioMove: Applying the Spatial Audio to Multi-Directional Limb Exercise Guidance. *Proceedings of the Association for Computing Machinery on Human-Computer Interaction* 8, MHCI, Article 244 (Sept. 2024), 26 pages. doi:10.1145/3676489
- [69] Lei Xiao, Yang Zhao, Dave Lindberg, Joel Hegland, Seth Moczydlowski, Eric Penner, Dan Tebbs, Daniel Terpstra, Ian Ender, Yu-Jen Lin, Julia Majors, and Douglas Lanman. 2025. Wide Field-of-View Mixed Reality. In *Association for Computing Machinery SIGGRAPH 2025 Emerging Technologies (SIGGRAPH '25)*. Association for Computing Machinery, New York, USA, Article 21, 2 pages. doi:10.1145/3721257.3734021
- [70] Soojeong Yoo and Judy Kay. 2017. Body-map: visualising exertion in virtual reality games. In *Proceedings of the 29th Australian Conference on Computer-Human Interaction (OzCHI '17)*. Association for Computing Machinery, New York, USA, 523–527. doi:10.1145/3152771.3156170
- [71] Xingyao Yu, Katrin Angerbauer, Peter Mohr, Denis Kalkofen, and Michael Sedlmair. 2020. Perspective matters: Design implications for motion guidance in mixed reality. In *2020 Institute of Electrical and Electronics Engineers international symposium on mixed and augmented reality (ISMAR)*. Institute of Electrical and Electronics Engineers, Piscataway, New Jersey, USA, 577–587.
- [72] Ye Yuan, Nathalie Riche, Nicolai Marquardt, Molly Jane Nicholas, Teddy Seyed, Hugo Romat, Bongshin Lee, Michel Pahud, Jonathan Goldstein, Rojin Vishkaie, Christian Holz, and Ken Hinckley. 2022. Understanding Multi-Device Usage Patterns: Physical Device Configurations and Fragmented Workflows. In *Proceedings of the 2022 CHI Conference on Human Factors in Computing Systems (CHI '22)*. Association for Computing Machinery, New York, USA, Article 64, 22 pages. doi:10.1145/3491102.3517702